

The Markov part of the shifted Weil transfer

A positive measure minus twice its exponential moving average: the decomposition, what can realize its pieces, and the contraction margin at the first horizon

Drafted for Edward Baker

Working manuscript, version 0.1
17 September 2026

Abstract

The shifted Weil family turns the Riemann hypothesis into a statement about a one-parameter family of causal all-pass filters: for each shift $\omega > 0$ the transfer $K_\omega(p) = \xi(\frac{1}{2} + p - \omega)/\xi(\frac{1}{2} + p + \omega)$ has unit modulus on the critical line, its compressions to windows of every length are contractions if and only if no zero lies further than ω from the line, and the first-order defect of the compression is the localized Weil form. A proposal to realize this flow as the deformation of a Wilson line, and a subsequent proposal to construct the deformation paths by a Loewner-type evolution, both ask the same question: what, exactly, would a realization have to produce? This manuscript answers it by factoring the transfer. Writing $\xi(u) = \frac{1}{2}u(u-1)\Lambda(u)$, the ratio $\Lambda(s-\omega)/\Lambda(s+\omega)$ is completely monotone, hence the Laplace transform of a positive measure — an explicit one, the convolution of a Beta-distributed continuous delay with a comb on the integer ratios whose weights are the Dirichlet coefficients of $\zeta(s-\omega)/\zeta(s+\omega)$ — and the polynomial ratio is a rational all-pass factor forced by unimodularity. Cleanly: $K_\omega = B_b \widehat{K}_\omega$ with B_b one Blaschke factor at the pole of ζ and $\widehat{K}_\omega$ completely monotone, so the causal kernel is *a positive measure minus twice its exponential moving average*. The Beta part is Bessel additivity in dimensions $\frac{1}{2} \mp \omega$; the comb is the Hecke semigroup of $\mathbb{N}^\times$ acting by dilations, which no Loewner chain produces in any reading of the variables; the correction carries no arithmetic; at $\omega = \frac{1}{2}$ the positive part is the Eisenstein scattering matrix of the modular surface and the comb weights are Euler's totient. None of this is a positivity mechanism — the subtraction cancels the growth of the positive part exactly and what remains is the fluctuation the zeros control — but it makes the transfer, its compression, its norm and its defect elementary functions of the integers below e^L . We build that assembly and validate it end to end at the first horizon $L = \log 3$: the compression is a strict contraction whose defect eigenvalue reproduces the margin of the Weil form in the same basis to 0.1% at $\omega = 0.01$; the exact defect expands as $2\omega Q - \omega^2(2Q^2 + [Q, A])$, whose commutator vanishes on the minimizer; the Galerkin excess is a truncation term that dies under nesting as predicted; and the first-order mass beyond the horizon has a closed form from ξ'/ξ alone that the assembled kernel matches to five digits. Nothing here constructs a source, proves positivity, or assumes the Riemann hypothesis except where a statement is labelled conditional.

1 Introduction

This manuscript reports the first results of the *Loewner* investigation of the SUSY-positivity program. It is written to be read on its own, and its introduction is deliberately slow: the results in Sections 3 to 5 are short and elementary once the objects are in view, and the work of the investigation was largely in getting them into view. The chain of logic runs from Weil's criterion, through the shifted family and its reading as a filter, through a proposal to realize that filter by a conformal evolution,

to a factorization of the filter that says what any realization would have to produce — and then to a computation that confirms the factorization is what it claims to be.

1.1 Weil's criterion, and where the arithmetic hides

Riemann's zeta function has a completed form $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, entire, real on the real axis, symmetric under $s \mapsto 1-s$, whose zeros are the nontrivial zeros ρ of ζ . The Riemann hypothesis places them all on the critical line $\text{Re } s = \frac{1}{2}$. Weil's criterion restates this as the positivity of a quadratic form. Take a test function f supported in a window $I_L = (-\frac{L}{2}, \frac{L}{2})$ of the real line, extend it by zero to F , and write $\gamma_\rho = -i(\rho - \frac{1}{2})$, so that γ_ρ is real exactly when ρ is on the line. The *Weil form* is

$$Q_L[f] = \sum_{\rho} \widehat{F}(\gamma_\rho) \overline{\widehat{F}(\gamma_\rho)}, \quad (1.1)$$

and the criterion says that RH holds if and only if $Q_L[f] \geq 0$ for every L and every f . Under RH each term is $|\widehat{F}(\gamma_\rho)|^2$ and the positivity is obvious; a zero off the line pairs with its reflection to produce a term that can be made negative by a suitable f .

The reason this is not a tautology is the explicit formula. The same quantity $Q_L[f]$ has a second expression in which the zeros do not appear at all: an *archimedean* term built from the digamma function, a *pole* term from the factor $s(s-1)$, and a *prime* term, $-2 \sum_{m \log p < L} (\log p) p^{-m/2} \text{Re} \langle F, U_{m \log p} F \rangle$, in which F is correlated with its own translates by the logarithms of prime powers. The prime-side expression is what one can compute, and what a construction could hope to produce; the zero-side expression is what is positive. The whole difficulty of the criterion is that positivity is manifest on one side and invisible on the other, and the two sides are the same number. The cancellation between them is extreme: at $L = \log 3$ the smallest value of Q_L on normalized f is about 5×10^{-8} , at $L = \log 7$ about 10^{-27} , while the individual terms are of order one. Every investigation of this program is, in one way or another, a search for a presentation of Q_L in which positivity is structural and the arithmetic is nevertheless present.

1.2 The shifted family: RH as the stability of a filter

The Wilson-lines investigation [2] studies a one-parameter deformation of the criterion. Write $s = \frac{1}{2} + p$ and, for $0 < \omega \leq \frac{1}{2}$,

$$K_\omega(p) = \frac{\xi(\frac{1}{2} + p - \omega)}{\xi(\frac{1}{2} + p + \omega)}. \quad (1.2)$$

Three facts organize everything that follows, and all three are elementary. First, K_ω is *unimodular on the imaginary axis*: the functional equation and the reality of ξ make numerator and denominator complex conjugates at $p = i\tau$, so $|K_\omega(i\tau)| = 1$ for every real τ . In the language of signal processing K_ω is an *all-pass filter*: it changes the phase of every frequency and the amplitude of none. Second, K_ω is the symbol of a *causal* operator: realized as a Laplace transform on a line $\text{Re } p = \eta$ to the right of all its singularities, it is $\int_0^\infty e^{-pt} k_\omega(t) dt$ for a kernel k_ω supported on $t \geq 0$, and the operator $V_\omega f = k_\omega * f$ never moves mass to the left. Third, the derivative at $\omega = 0$ is the Weil form. Compress V_ω to the window, $V_{\omega,L} = P_L V_\omega P_L$ with P_L the restriction to I_L , and form the *contraction defect* $D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}$, which measures how far the compression is from an isometry. Then

$$\|f\|^2 - \|V_{\omega,L} f\|^2 = \langle f, D_{\omega,L} f \rangle = 2\omega Q_L[f] + O(\omega^2). \quad (1.3)$$

So Weil positivity on I_L is the statement that, to first order in the shift, the compressed filter contracts.

The three facts combine into a dichotomy [2, Proposition 7.2] that is the hinge of the present manuscript. Fix ω . On the whole line an all-pass filter is unitary, and a compression of a unitary is a contraction; but V_ω is unitary on $L^2(\mathbb{R})$ only if the causal realization can be moved onto the imaginary axis, which requires K_ω to have no poles in the right half-plane. The poles of K_ω are at $p = \rho - \frac{1}{2} - \omega$, so this is the statement that no zero has $\text{Re } \rho > \frac{1}{2} + \omega$. If there is such a zero, k_ω contains a growing exponential and the compressions are unbounded as L grows. Hence

$$\sup_L \|V_{\omega,L}\| \in \{1, \infty\}, \quad \sup_L \|V_{\omega,L}\| = 1 \iff \text{no zero of } \xi \text{ lies further than } \omega \text{ from the line.} \quad (1.4)$$

The shifted family is a graded criterion, one zero-free strip per shift, and the Riemann hypothesis is the statement that, for every $\omega > 0$, the causal filter k_ω is *passive*: every window compression is a contraction. Under RH the defect has a physical meaning, the flux identity [2, Proposition 4.5]: $\langle f, D_{\omega,L} f \rangle$ is exactly the mass of $V_\omega f$ that has been pushed past the leading end of the window. The form Q_L is the first-order version of that flux.

1.3 Realizations, and the two proposals

The bulk-realization program asks for a system — a field theory, a statistical model, a conformal evolution — in which the transfer, or the form, appears as an observable whose positivity is structural, so that the arithmetic would be carried by the system and its positivity by a general theorem (unitarity, reflection positivity, subordination). The dichotomy sets the standard any such realization must meet: it must be exactly critical, unitary on the whole line and contracting on every window by a margin ωm_L that collapses superexponentially in L [2, Section 8].

The Wilson-lines proposal [2] took the transfer to be the deformation flow of an open Wilson line, because the shifted family obeys a flow equation $\partial_\omega V_{\omega,L} = -G_{\omega,L} V_{\omega,L}$ of the shape produced by varying a line's contour, with the non-Abelian Stokes theorem supplying the coefficient. Two of that manuscript's findings bear directly on what follows. The shift direction is *abelian*: compression to a window is multiplicative on causal kernels, so the transfers and their generators lie in one commutative algebra — the algebra of translation-invariant causal operators on the line, lower-triangular Toeplitz matrices in any discretization — and what is abelian is precisely translation invariance along the arithmetic coordinate [2, Theorem 5.2]. And the two-parameter flow in the shift-length plane is flat [2, Proposition 9.4]. The inherited note [4] from which this investigation was opened sharpened both statements, in response to the question of whether these were obstructions or selections. They are selections. Flatness is a tautology for any single-valued family $V(\alpha)$ with $dV = \mathcal{A}V$ — the curvature of $\mathcal{A} = dV V^{-1}$ vanishes identically — so it constrains nothing about a realization; in the non-Abelian Stokes variation the field strength is the *coefficient* $\mathcal{A}$, not the curvature, and the object to match is the generator $G_{\omega,L}$ itself. Commutativity, likewise, is a symmetry requirement on the insertions that implement the shift (they must act as functions of one operator, uniformly along the line), not an impossibility. Along a general path in the (ω, L) plane the transport is a genuinely ordered product, because the endpoint generator B_L and the bulk generator $G_{\omega,L}$ generate a triangular, not commutative, algebra.

The Loewner proposal, made at that point, was to construct the paths independently: to take the arithmetic coordinate $x = \log r$ as the time of a whole-plane Loewner evolution, whose hull grows from 0 toward ∞ with logarithmic capacity t , and to let a simple boundary driving function produce the ordered, infinite-dimensional, structurally contractive family that the shift direction lacks. The affinities are real: Loewner time is log-conformal-radius, which is the arithmetic coordinate exactly; the marked points 0, ∞ have the dilations and inversions of the Weil form's exact symmetry group as their stabilizer; the Loewner equation is a tip equation with one scalar coefficient, like the endpoint

variation of the transfer; and composition with univalent self-maps is contractive on Hardy-type spaces by Littlewood subordination, with a positive defect and no arithmetic in it.

1.4 What a realization would have to produce: the factorization

To test either proposal one must know the kernel k_ω concretely, and it turns out to have a structure that neither proposal anticipated. Factor the completed zeta function as $\xi(u) = \frac{1}{2}u(u-1)\Lambda(u)$ with $\Lambda(u) = \pi^{-u/2}\Gamma(u/2)\zeta(u)$. The transfer splits correspondingly into a ratio of Λ 's and a ratio of quadratics, $K_\omega = R_\omega \tilde{K}_\omega$, and the two factors have opposite characters.

The ratio $\tilde{K}_\omega(p) = \Lambda(s-\omega)/\Lambda(s+\omega)$ is *completely monotone* on $p > \frac{1}{2} + \omega$: all its derivatives alternate in sign. By Bernstein's theorem a completely monotone function is the Laplace transform of a positive measure, so the causal kernel $\tilde{k}_\omega$ of $\tilde{K}_\omega$ is *nonnegative*. Moreover it is explicit. The Γ -ratio $\pi^\omega \Gamma(\frac{s-\omega}{2})/\Gamma(\frac{s+\omega}{2})$ is a Beta integral whose kernel is $k_\omega^\Gamma(x) = \frac{2\pi^\omega}{\Gamma(\omega)}(2\sinh x)^{\omega-1}e^{x/2}$, a continuous delay distributed as $x = -\frac{1}{2}\log U$ with $U \sim \text{Beta}(\frac{1}{4} - \frac{\omega}{2}, \omega)$; and the ζ -ratio $\zeta(s-\omega)/\zeta(s+\omega)$ is a Dirichlet series whose coefficients $\tilde{c}_n = n^{\omega-1/2} \prod_{p|n}(1-p^{-2\omega})$ are positive because the Euler product makes each local factor a geometric series with positive terms. Its kernel is a comb of atoms at the integer logarithms, and $\tilde{k}_\omega = k_\omega^\Gamma * \sum_n \tilde{c}_n \delta_{\log n}$: a sum of Beta-shaped spikes, one at every $\log n$, with positive weights. This is the *Markov part* of the transfer, a positive kernel, though of infinite mass — it grows like $e^{(\frac{1}{2}+\omega)x}$, because $\Lambda(s-\omega)$ has the pole of ζ at $s-\omega=1$, and that pole is the prime number theorem's main term.

The rational factor R_ω is a ratio of two Blaschke factors, unimodular on the axis, and it is *forced*: given the Markov part, it is the unique minimal rational all-pass function that cancels the poles and zeros of $\tilde{K}_\omega$ at the poles of Λ , so its coefficients carry nothing but the positions 0, 1 of those poles and the shift. Absorbing the harmless half of it into the Markov part gives the form we use throughout:

$$K_\omega = B_b \widehat{K}_\omega, \quad B_b(p) = \frac{p-b}{p+b}, \quad b = \frac{1}{2} + \omega, \quad \widehat{K}_\omega \text{ completely monotone on } p > b, \quad (1.5)$$

and at the level of kernels

$$k_\omega = \widehat{k}_\omega - 2 \text{EMA}_b[\widehat{k}_\omega], \quad \text{EMA}_b[g](x) = b \int_0^x e^{-b(x-y)} g(y) dy, \quad \widehat{k}_\omega \geq 0. \quad (1.6)$$

The impulse response of the shifted Weil transfer is a positive measure minus twice its exponential moving average, at the rate set by the pole of ζ . The subtraction is exact and unconditional: it maps the main term Ce^{bx} of the positive part to Ce^{-bx} , so the growth is cancelled identically, and what remains of k_ω is the fluctuation of $\widehat{k}_\omega$ about its main term — which is to say, the zeros. Under RH the remainder decays like $e^{-\omega x}$; a zero at distance $\delta > \omega$ from the line leaves a growing $e^{(\delta-\omega)x}$. This is the dichotomy (1.4) at the level of the kernel, and it says where the arithmetic hides: not in the small rational correction, whose residues are fixed by the functional equation, but in the fine structure of a positive measure whose Laplace transform has poles at the zeros behind its abscissa of convergence.

1.5 What the pieces are, and what can make them

Each piece of the positive part has a name in probability or geometry, and the names decide the two proposals. The Beta law is *Bessel additivity*: with $a = \frac{1}{2} - \omega$ and $b = \frac{1}{2} + \omega$, $U = Z_a/(Z_a + Z_{2\omega})$ for independent squared Bessel processes of dimensions a and 2ω , whose sum is a squared Bessel process of dimension b . The two exponents of the rational factor are the two Bessel dimensions, and

the delay test's archimedean quarter is half a Bessel dimension. This is a statement about Bessel processes, which drive radial Loewner chains but are not themselves chains; whether a radial chain has this law as a squared-radius ratio is the one question in the Loewner direction that remains open.

The comb cannot come from a Loewner chain in any reading of the variables. If the arithmetic coordinate is Loewner time, the only translation-invariant filter a rotation-invariant chain induces on analytic observables is the pure delay e^{-tp} — a rotation average of a composition operator is a dilation. If the arithmetic coordinate is the radial coordinate $\log |z|$, the chain's transport of exterior points is a pure delay in the far field, $\log |g_t(z)| = \log |z| - t + O(|z|^{-1})$, and translation invariance forces the induced kernel to equal its far-field limit. Echoes at the fixed separations $\log n$ with position-independent weights are available in neither reading. What the comb *is* is a semigroup structure: the operator $C_\omega = \sum_n \tilde{c}_n \mu_n$ with $\mu_n f(x) = f(x - \log n)$ and $\mu_n \mu_m = \mu_{nm}$, a Dirichlet series in the multiplicative semigroup $\mathbb{N}^\times$ acting on the ray by dilations — the Hecke operator of the Bost–Connes system, with coefficients that are, to first order in ω , $2\omega \Lambda(n) n^{-1/2}$: the von Mangoldt function.

And at the endpoint $\omega = \frac{1}{2}$ the whole positive part is a classical geometric object: $\widehat{K}_{1/2} = \Lambda(p)/\Lambda(p+1)$ is the Eisenstein scattering matrix of the modular surface $PSL(2, \mathbb{Z}) \backslash \mathbb{H}$, the comb weights are Euler's totient $\varphi(n)/n$ — the coset count in the cusp's Fourier expansion that produces $\zeta(2s-1)/\zeta(2s)$ in the first place — and $K_{1/2} = B_1 \widehat{K}_{1/2}$ is that scattering matrix with its pole at $s=1$ removed by one Blaschke factor, which is exactly the Lax–Phillips modification. So the Markov part has, at one shift, a realization in which the comb is a coset sum and the correction is the removal of the residual spectrum. For $\omega < \frac{1}{2}$ no Eisenstein construction produces the offset 2ω , and the interior of the family has no such realization from that source.

1.6 What the decomposition is not, and what it is

It is not a positivity mechanism, and the dichotomy says why with no computation. The positive part $\widehat{K}_\omega$ has a causal operator $\widehat{V}_\omega$ of norm greater than one on every window (its kernel grows), the Blaschke factor is a contraction, and their product is the transfer; the contraction of $V_{\omega,L}$ for all L is a zero-free strip by (1.4), so no structural property of $\widehat{k}_\omega$ short of the location of the poles can imply it. Any passivity theorem for filters of the form “positive measure minus twice its EMA” would have to see the poles. We record this plainly because it is the honest end of one line of thought: positivity of the Markov part is real, explicit and cheap, and it proves nothing about the zeros.

What the decomposition is, is three things. It is a *computational tool*: on a window of length L only the atoms $\log n < L$ contribute, so the transfer, its compression, its norm and its defect are assembled exactly from $(2 \sinh x)^{\omega-1} e^{x/2}$, the numbers $\tilde{c}_n$ for $n < e^L$, and two exponentials — no ξ , no ζ on a line, no zeros — in precisely the way the Weil form is assembled from the explicit formula. It is a *dictionary*: Bessel additivity for the archimedean factor, the Hecke semigroup for the comb, one Blaschke factor at the pole of ζ for the correction, the modular surface at the endpoint. And it is a statement of the program's criticality in filter language: the shifted Weil transfer is a lossless filter whose impulse response is a positive measure minus twice its exponential moving average, and the Riemann hypothesis is the statement that this filter is stable at every $\omega > 0$.

1.7 The first computation

The tool was then built and tested. Section 5 assembles $V_{\omega,L}$ at the first horizon $L = \log 3$ in an orthonormal sine basis, from the kernel alone, and measures its contraction data. The compression is a strict contraction at every shift computed, and the smallest eigenvalue of its defect satisfies

$\lambda_{\min}(D_{\omega,L})/2\omega \rightarrow m_L$, the margin of the Weil form in the same basis, with agreement to 0.1% at $\omega = 0.01$: the transfer side and the form side of the program, assembled from different presentations of the explicit formula, agree at first order. The deviations at larger ω are then accounted for exactly. On the exact compression the defect expands as $D_{\omega,L} = 2\omega Q_L - \omega^2(2Q_L^2 + [Q_L, A_L]) + O(\omega^3)$, where A_L is the skew part of the compressed generator, and on the minimizer of Q_L the commutator term vanishes, so the only first-order correction to (1.3) at the minimizer is $-\omega m_L^2$. The positive excess seen in a finite basis is a truncation term, the energy of $A_L f$ outside the basis, and it dies under nesting exactly as predicted. Finally the mass of the assembled kernel beyond the horizon has, to first order in ω , a closed form from the logarithmic derivative of ξ alone — poles, digamma and primes, no zeros — and the assembly matches it to five digits.

1.8 Organization and status

Section 2 fixes conventions and quotes, with proofs where they are short, the facts about the shifted family that the rest uses. Section 3 proves the factorization and its reading. Section 4 identifies the pieces and proves the negative Loewner results. Section 5 presents the assembly, the second-order theory of the defect, the first-order tail, and the numerical results. Section 6 ranks what remains. Appendix A documents the programmes and records and the numerical lessons; Appendix B records provenance and the use of language models. Every statement is labelled as a written proof, a labelled numerical computation, or a reading; nothing assumes the Riemann hypothesis except where a statement says so.

Contents

1	Introduction	1
1.1	Weil's criterion, and where the arithmetic hides	2
1.2	The shifted family: RH as the stability of a filter	2
1.3	Realizations, and the two proposals	3
1.4	What a realization would have to produce: the factorization	4
1.5	What the pieces are, and what can make them	4
1.6	What the decomposition is not, and what it is	5
1.7	The first computation	5
1.8	Organization and status	6
2	Conventions, and the facts we use	8
2.1	The Weil form	8
2.2	The shifted family	8
2.3	Facts about the transfer	9
2.4	Complete monotonicity, Bernstein functions, Blaschke factors	10
3	The decomposition of the transfer	10
3.1	Factorization	10
3.2	The Λ -ratio is completely monotone	11
3.3	The rational factor is forced, and one all-pass section suffices	12
3.4	Reading	12
4	What the pieces are, and what can realize them	13
4.1	The two proposals, stated precisely	13
4.2	The archimedean factor is Bessel additivity	14
4.3	A Loewner chain cannot produce the comb	14
4.4	What the comb is	15
4.5	The endpoint $\omega = \frac{1}{2}$ is the modular surface	15
4.6	Summary of the dictionary	16
5	The contraction margin at the first horizon	16
5.1	The assembly	16
5.2	What the numbers should be	17
5.3	Results	18
5.4	Reading	20
6	What remains, ranked	20
6.1	Closed	20
6.2	Open	20
6.3	Status of every statement	21
A	Programmes, records, and numerical lessons	21
B	Provenance, and the use of language models	22
	References	24

2 Conventions, and the facts we use

2.1 The Weil form

Inner products are conjugate-linear in the first argument. For $L > 0$ let $I_L = (-L/2, L/2)$, let P_L be multiplication by $\mathbf{1}_{I_L}$, and for $f \in L^2(I_L)$ let F be its extension by zero. Write

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad U_d F(x) = F(x - d), \quad \|F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau, \quad (2.1)$$

and, for $r > 0$,

$$n_\gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}} = \frac{e^{r/2}}{2 \sinh r}. \quad (2.2)$$

The localized Weil form is, in the normalization of [2],

$$Q_L[f] = \int_0^\infty [2\|F\|^2 - 2\operatorname{Re}\langle F, U_r F \rangle] n_\gamma(r) dr + w_0 \|f\|^2 + P_L[f] - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re}\langle F, U_{m \log p} F \rangle, \quad (2.3)$$

with $w_0 = \psi(\frac{1}{4}) - \log \pi$, $\psi = \Gamma'/\Gamma$, and the pole form $P_L[f] = 2|\int f \cosh \frac{x}{2}|^2 - 2|\int f \sinh \frac{x}{2}|^2$. Its spectral form is

$$Q_L[f] = \sum_{\rho} \widehat{F}(\gamma_\rho) \overline{\widehat{F}(\overline{\gamma_\rho})}, \quad \gamma_\rho = -i(\rho - \tfrac{1}{2}), \quad (2.4)$$

the sum over the nontrivial zeros of ζ in symmetric order, and Weil's criterion, used as input, is $\text{RH} \iff Q_L \geq 0$ for every L . We write m_L for the smallest eigenvalue of the (unbounded below-bounded) operator representing Q_L on $L^2(I_L)$, the *margin*; under RH it is positive [2, Section 8], and its recorded values at $L = \log 3, \log 5, \log 7$ are $5.5 \times 10^{-8}, 9.3 \times 10^{-18}, 6.8 \times 10^{-28}$ [7, 8].

2.2 The shifted family

Let

$$\Lambda(u) = \pi^{-u/2} \Gamma(u/2) \zeta(u), \quad \xi(u) = \tfrac{1}{2} u(u-1) \Lambda(u), \quad s = \tfrac{1}{2} + p, \quad (2.5)$$

so that ξ is entire, $\xi(u) = \xi(1-u)$, ξ is real on the real axis, and Λ has simple poles at $u = 0, 1$ with residues $-1, 1$ and is otherwise holomorphic except that it inherits nothing from Γ (the trivial zeros of ζ cancel the poles of $\Gamma(u/2)$). Throughout,

$$0 < \omega \leq \tfrac{1}{2}, \quad a = \tfrac{1}{2} - \omega, \quad b = \tfrac{1}{2} + \omega, \quad (2.6)$$

and the transfer symbol is

$$K_\omega(p) = \frac{\xi(s - \omega)}{\xi(s + \omega)}. \quad (2.7)$$

The poles of K_ω are at the zeros of $\xi(s + \omega)$, $p = \rho - \frac{1}{2} - \omega$, all with $\operatorname{Re} p \leq \frac{1}{2} - \omega = a$; its zeros are at $p = \rho - \frac{1}{2} + \omega$. The *causal realization* of a symbol H holomorphic on $\operatorname{Re} p > \eta_0$ is the convolution with its inverse Laplace transform on any line $\operatorname{Re} p = \eta > \eta_0$, a distribution h supported on $[0, \infty)$ with $H(p) = \int_0^\infty e^{-pt} h(t) dt$; we write $V_\omega f = k_\omega * f$ for K_ω , and

$$V_{\omega,L} = P_L V_\omega P_L, \quad D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}, \quad (2.8)$$

for the compression to the window and its contraction defect. Causal distributions with a Laplace transform form a commutative convolution algebra (the algebra $\mathcal{D}'_+$ of Schwartz), on which the Laplace transform is a homomorphism; we use this freely.

The *generator* is the causal realization $G_{\omega,L} = P_L(g_\omega * \cdot)P_L$ of the symbol

$$a_\omega(p) = -\partial_\omega \log K_\omega(p) = \frac{\xi'}{\xi}(s+\omega) + \frac{\xi'}{\xi}(s-\omega), \quad \frac{\xi'}{\xi}(u) = \frac{1}{u} + \frac{1}{u-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{u}{2}\right) + \frac{\zeta'}{\zeta}(u), \quad (2.9)$$

and the *shifted form* is its Hermitian part, $Q_{\omega,L}[g] = \text{Re}\langle g, G_{\omega,L}g \rangle$, with $Q_{0,L} = Q_L$. The flow identities [2, Section 2] are

$$\partial_\omega V_{\omega,L} = -G_{\omega,L}V_{\omega,L}, \quad V_{0,L} = I, \quad \frac{d}{d\omega} \|V_{\omega,L}f\|^2 = -2Q_{\omega,L}[V_{\omega,L}f], \quad (2.10)$$

so that $\langle f, D_{\omega,L}f \rangle = 2\omega Q_L[f] + O(\omega^2)$, which is (1.3). Two structural facts about a_ω are used below. It is *even in* ω , by inspection of (2.9); at the kernel level, since a shift of p multiplies a kernel by an exponential, $g_\omega(t) = 2 \cosh(\omega t) g^+(t)$ with g^+ the kernel of $\frac{\xi'}{\xi}(\frac{1}{2} + p)$, which is [2, Theorem 3.1]: the shifted form is the central form with every off-diagonal kernel and every prime atom multiplied by $\cosh(\omega \cdot)$. And its kernel at $\omega = 0$ is read off term by term from (2.9):

$$g_0(t) = 2e^{-t/2} + 2e^{t/2} - \log \pi \delta_0 - \text{fp}[2n_\gamma] - 2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta_{\log n}, \quad (2.11)$$

where the finite part is the causal distribution with Laplace transform $-\psi(\frac{1}{4} + \frac{p}{2})$, fixed by the integral representation $\psi(z) = \int_0^\infty (\frac{e^{-t}}{t} - \frac{e^{-zt}}{1-e^{-t}}) dt$:

$$\langle \text{fp}[2n_\gamma], \phi \rangle = \lim_{\epsilon \downarrow 0} \left[\int_\epsilon^\infty 2n_\gamma(u) \phi(u) du - \phi(0) E_1(2\epsilon) \right], \quad E_1(x) = \int_x^\infty \frac{e^{-u}}{u} du. \quad (2.12)$$

(Here g_0 is twice ξ'/ξ at $u = \frac{1}{2} + p$: the constant $-\frac{1}{2} \log \pi$ doubles to $-\log \pi \delta_0$, and the pole terms $\frac{1}{u} + \frac{1}{u-1} = \frac{1}{p+1/2} + \frac{1}{p-1/2}$ have kernels $e^{-t/2}$ and $e^{t/2}$, doubled.)

2.3 Facts about the transfer

The following are proved in [2]; we reprove the two that are one line, state the rest, and use nothing else from that manuscript.

Proposition 2.1 (Unimodularity [2, Proposition 4.1]). $|K_\omega(i\tau)| = 1$ for every real τ with $\xi(\frac{1}{2} + \omega + i\tau) \neq 0$.

Proof. $\xi(u) = \xi(1-u)$ gives $\xi(\frac{1}{2} - \omega + i\tau) = \xi(\frac{1}{2} + \omega - i\tau)$, and $\xi(\bar{u}) = \overline{\xi(u)}$ gives $\xi(\frac{1}{2} + \omega - i\tau) = \overline{\xi(\frac{1}{2} + \omega + i\tau)}$; numerator and denominator of (2.7) are conjugate. $\square$

Lemma 2.2 (Compression is multiplicative on causal kernels [2, Lemma 5.1]). *If $\mathcal{A}, \mathcal{B}$ are causal convolutions and f is supported in I_L , then $P_L \mathcal{A} \mathcal{B} P_L f = (P_L \mathcal{A} P_L)(P_L \mathcal{B} P_L) f$. Consequently $H \mapsto P_L(h * \cdot)P_L$ is a homomorphism from the causal convolution algebra onto a commutative algebra $\mathcal{A}_L$ of operators on $L^2(I_L)$, in which all $V_{\omega,L}$ and $G_{\omega,L}$ lie.*

Proof. $(I - P_L)\mathcal{B}f$ is supported to the right of I_L ; a causal operator moves support only to the right; so $P_L \mathcal{A}(I - P_L)\mathcal{B}P_L f = 0$. $\square$

Proposition 2.3 (Flux and dichotomy [2, Propositions 4.5, 7.1, 7.2]). *For fixed ω , $L \mapsto \|V_{\omega,L}\|$ is nondecreasing with limit $\|V_{\omega}\|_{L^2(\mathbb{R})}$. If no zero of ξ has $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$, then K_{ω} is inner on the right half-plane, V_{ω} is unitary and causal on $L^2(\mathbb{R})$, every $V_{\omega,L}$ is a contraction, and*

$$\langle f, D_{\omega,L} f \rangle = \|(I - P_L)V_{\omega} f\|^2 = \int_{L/2}^{\infty} |(V_{\omega} f)(x)|^2 dx. \quad (2.13)$$

If some zero has $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$, then $\sup_L \|V_{\omega,L}\| = \infty$.

Sketch. Nested compressions; strong convergence $P_L \rightarrow I$. If no pole of K_{ω} lies in $\operatorname{Re} p > 0$ the realization contour may be moved to the imaginary axis, where K_{ω} is unimodular, so V_{ω} is unitary and $\|f\|^2 - \|P_L V_{\omega} f\|^2 = \|(I - P_L)V_{\omega} f\|^2$, with the escaping mass to the right by causality. A pole at $\operatorname{Re} p = \delta - \omega > 0$ contributes a growing exponential to k_{ω} , making V_{ω} unbounded, hence the compressions unbounded in L . $\square$

2.4 Complete monotonicity, Bernstein functions, Blaschke factors

A function H on an interval (p_0, ∞) is *completely monotone* if $(-1)^k H^{(k)} \geq 0$ for all $k \geq 0$. Bernstein's theorem states that H is completely monotone on (p_0, ∞) if and only if $H(p) = \int_{[0, \infty)} e^{-pt} d\mu(t)$ for a positive measure μ with the integral finite for $p > p_0$; the measure is the causal inverse Laplace kernel of H . Products of completely monotone functions are completely monotone, and if Φ has a completely monotone derivative on (p_0, ∞) then $e^{-\Phi}$ is completely monotone there (after adding a constant, Φ is a Bernstein function, and $e^{-\Phi}$ is then completely monotone by the standard composition theorem; see [9]). Both facts are used in Section 3.

A *Blaschke factor* of the right half-plane is $B_c(p) = \frac{p-c}{p+\bar{c}}$ with $\operatorname{Re} c > 0$; it is holomorphic and bounded on $\operatorname{Re} p > 0$, unimodular on the axis, and its causal kernel is $\delta_0 - 2\operatorname{Re} c e^{-\bar{c}t} \mathbf{1}_{t>0}$. A finite product of Blaschke factors is the general rational function unimodular on the axis and equal to 1 at infinity. The *exponential moving average* at rate $b > 0$ is $\operatorname{EMA}_b[g](x) = b \int_0^x e^{-b(x-y)} g(y) dy$, the causal convolution with $b e^{-bt} \mathbf{1}_{t>0}$, of symbol $\frac{b}{p+b}$; so $B_b = 1 - \frac{2b}{p+b}$ acts on kernels as $g \mapsto g - 2\operatorname{EMA}_b[g]$.

3 The decomposition of the transfer

Everything in this section is unconditional and elementary. The plan is to split ξ into Λ and the polynomial, show that the Λ -ratio is completely monotone and compute its positive kernel in closed form, show that the polynomial ratio is a forced rational all-pass factor, and then read the result.

3.1 Factorization

Proposition 3.1 (Factorization). $K_{\omega} = R_{\omega} \widetilde{K}_{\omega}$ with

$$\widetilde{K}_{\omega}(p) = \frac{\Lambda(s - \omega)}{\Lambda(s + \omega)}, \quad R_{\omega}(p) = \frac{(s - \omega)(s - \omega - 1)}{(s + \omega)(s + \omega - 1)} = \frac{(p + a)(p - b)}{(p + b)(p - a)} = 1 - \frac{4\omega b}{p + b} - \frac{4\omega a}{p - a}. \quad (3.1)$$

Apart from its poles at the zeros of $\Lambda(s + \omega)$ and its zeros at the zeros of $\Lambda(s - \omega)$, $\widetilde{K}_{\omega}$ has exactly two poles, at $p = b$ and $p = -a$ (the poles of $\Lambda(s - \omega)$ at $s - \omega = 1, 0$), and exactly two zeros, at $p = a$ and $p = -b$ (the poles of $\Lambda(s + \omega)$); R_{ω} cancels all four. Both factors are unimodular on $\operatorname{Re} p = 0$.

Proof. $\xi(u) = \frac{1}{2}u(u-1)\Lambda(u)$ at $u = s \mp \omega$; $s - \omega = p + a$, $s - \omega - 1 = p - b$, $s + \omega = p + b$, $s + \omega - 1 = p - a$. For the partial fractions, $(p+a)(p-b) - (p+b)(p-a) = 2(a-b)p = -4\omega p$, so $R_\omega - 1 = \frac{-4\omega p}{(p+b)(p-a)}$, whose residues at $p = -b$ and $p = a$ are $-4\omega b$ and $-4\omega a$. $R_\omega = B_b/B_a$ is a ratio of two Blaschke factors, hence unimodular on the axis; $\widetilde{K}_\omega = K_\omega/R_\omega$ is then unimodular by Proposition 2.1. $\square$

3.2 The Λ -ratio is completely monotone

Proposition 3.2 (Complete monotonicity). *$\widetilde{K}_\omega$ is completely monotone on (b, ∞) . Hence its causal kernel $\widetilde{k}_\omega$ is a positive measure on $[0, \infty)$ with $\int_0^\infty e^{-pt} \widetilde{k}_\omega(t) dt < \infty$ for $p > b$.*

Proof. On $p > b$ both $\Lambda(s \pm \omega)$ are positive ($s - \omega > 1$), so $\Phi(p) = -\log \widetilde{K}_\omega(p) = \log \Lambda(s + \omega) - \log \Lambda(s - \omega)$ is real, and with $\frac{\Lambda'}{\Lambda}(u) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(\frac{u}{2}) + \frac{\zeta'}{\zeta}(u)$,

$$\Phi'(p) = \int_{-\omega}^{\omega} \left[\frac{1}{4} \psi' \left(\frac{s+t}{2} \right) + \sum_{n \geq 2} \Lambda(n) \log n n^{-(s+t)} \right] dt, \quad (3.2)$$

since $\frac{d}{du} \frac{\zeta'}{\zeta}(u) = \sum \Lambda(n) \log n n^{-u}$ for $\operatorname{Re} u > 1$ and $s+t \geq s-\omega > 1$ on the range. Each integrand is completely monotone in p : $\psi'(z) = \sum_{k \geq 0} (z+k)^{-2}$ is, and n^{-p} is. So Φ' is completely monotone on (b, ∞) , and $e^{-\Phi} = \widetilde{K}_\omega$ is completely monotone there (Section 2). Bernstein's theorem gives the measure. $\square$

Proposition 3.3 (The pieces). *$\widetilde{K}_\omega = K_\omega^\Gamma K_\omega^\zeta$ with*

$$K_\omega^\Gamma(p) = \pi^\omega \frac{\Gamma(\frac{s-\omega}{2})}{\Gamma(\frac{s+\omega}{2})} = \int_0^\infty e^{-px} k_\omega^\Gamma(x) dx, \quad k_\omega^\Gamma(x) = \frac{2\pi^\omega}{\Gamma(\omega)} (2 \sinh x)^{\omega-1} e^{x/2}, \quad (3.3)$$

$$K_\omega^\zeta(p) = \frac{\zeta(s-\omega)}{\zeta(s+\omega)} = \sum_{n \geq 1} \widetilde{c}_n e^{-p \log n}, \quad \widetilde{c}_n = n^{\omega-\frac{1}{2}} \prod_{p|n} (1 - p^{-2\omega}) > 0, \quad (3.4)$$

so that

$$\widetilde{k}_\omega = k_\omega^\Gamma * \sum_{n \geq 1} \widetilde{c}_n \delta_{\log n} = \sum_{n \geq 1} \widetilde{c}_n k_\omega^\Gamma(\cdot - \log n) \geq 0, \quad (3.5)$$

a sum of shifted Beta kernels with positive weights; note $k_\omega^\Gamma = \frac{2\pi^\omega}{\Gamma(\omega)} (2 \sinh x)^\omega n_\gamma(x)$ by (2.2). Moreover $k_\omega^\Gamma(x) dx$ is the law of $x = -\frac{1}{2} \log U$ with $U \sim \operatorname{Beta}(\frac{a}{2}, \omega)$, up to the factor $\pi^\omega \Gamma(\frac{a}{2}) / \Gamma(\frac{b}{2})$:

$$K_\omega^\Gamma(p) = \pi^\omega \frac{\Gamma(\frac{a}{2})}{\Gamma(\frac{b}{2})} \mathbb{E}[U^{p/2}]. \quad (3.6)$$

Proof. The Beta integral $\int_0^1 r^{s-\omega-1} (1-r^2)^{\omega-1} dr = \frac{1}{2} B(\frac{s-\omega}{2}, \omega) = \frac{\Gamma(\frac{s-\omega}{2}) \Gamma(\omega)}{2\Gamma(\frac{s+\omega}{2})}$, with $r = e^{-x}$, is $\int_0^\infty e^{-(s-\omega)x} (1-e^{-2x})^{\omega-1} dx$; with $s = \frac{1}{2} + p$ and $1-e^{-2x} = e^{-x} 2 \sinh x$, the integrand is $e^{-px} (2 \sinh x)^{\omega-1} e^{x/2}$, which gives (3.3). For (3.4): $\zeta(s-\omega) = \sum n^\omega n^{-s}$ and $1/\zeta(s+\omega) = \sum \mu(n) n^{-\omega} n^{-s}$, so the coefficient c_n of n^{-s} is multiplicative with $c_{p^k} = p^{k\omega} - p^{(k-2)\omega} = p^{k\omega} (1 - p^{-2\omega})$ for $k \geq 1$; writing $n^{-s} = n^{-1/2} e^{-p \log n}$ puts the weight $\widetilde{c}_n = c_n n^{-1/2}$ on the atom at $\log n$. Positivity: $p^{-2\omega} < 1$. Convolution theorem for (3.5). For (3.6): $\mathbb{E} U^{p/2} = B(\frac{a}{2} + \frac{p}{2}, \omega) / B(\frac{a}{2}, \omega)$ and $\frac{a}{2} + \frac{p}{2} = \frac{s-\omega}{2}$, $\frac{a}{2} + \omega = \frac{b}{2}$, $\frac{a}{2} + \omega + \frac{p}{2} = \frac{s+\omega}{2}$. $\square$

Two features of $\tilde{k}_\omega$ matter below. Near each atom the spike is an integrable power, $k_\omega^\Gamma(\tau) \sim \frac{2\pi^\omega}{\Gamma(\omega)}(2\tau)^{\omega-1}$ as $\tau \downarrow 0$, so $\tilde{k}_\omega$ has no atoms and is not bounded; as $\omega \downarrow 0$, $k_\omega^\Gamma \rightarrow \delta_0$ weakly while spreading its mass as $\tau^{\omega-1}$, so that for small ω half the mass sits below $\tau = 2^{-1/\omega}$. And $\tilde{k}_\omega$ grows: the pole of $\tilde{K}_\omega$ at $p = b$ makes $\tilde{k}_\omega(x) \sim C_\omega e^{bx}$ on average, with C_ω the residue there.

3.3 The rational factor is forced, and one all-pass section suffices

Proposition 3.4 (Forcedness). *Among rational functions ρ with $|\rho(i\tau)| = 1$ and $\rho(\infty) = 1$, the unique one of minimal degree such that $\rho\tilde{K}_\omega$ is holomorphic and nonvanishing at the four points $p \in \{b, -a, a, -b\}$ is $\rho = R_\omega$.*

Proof. Such ρ is a product of Blaschke factors $\frac{p-c}{p+\bar{c}}$. Cancelling the pole of $\tilde{K}_\omega$ at b needs the factor $\frac{p-b}{p+\bar{b}}$, whose own pole at $-b$ is cancelled by the zero of $\tilde{K}_\omega$ there; cancelling the pole at $-a$ needs $\frac{p+a}{p-\bar{a}}$, whose pole at a is cancelled by the zero there. Any further factor introduces an uncanceled pole or zero. $\square$

So, given the Markov part, the correction is determined by the functional equation; its residues $-4\omega b$, $-4\omega a$ are not data a realization could get right or wrong. A realization that produced $\tilde{K}_\omega$ and respected unimodularity would produce K_ω . The arithmetic is in the poles of $\tilde{K}_\omega$ at the zeros of $\Lambda(s + \omega)$ and nowhere else.

Proposition 3.5 (One all-pass section). *With $B_b(p) = \frac{p-b}{p+\bar{b}}$ and*

$$\widehat{K}_\omega(p) = \frac{p+a}{p-a} \tilde{K}_\omega(p) = \frac{(s-\omega)\Lambda(s-\omega)}{(s+\omega-1)\Lambda(s+\omega)}, \quad (3.7)$$

one has $K_\omega = B_b \widehat{K}_\omega$, and $\widehat{K}_\omega$ is completely monotone on (b, ∞) . Hence $\widehat{k}_\omega \geq 0$ and

$$k_\omega = \widehat{k}_\omega - 2 \text{EMA}_b[\widehat{k}_\omega], \quad \widehat{k}_\omega = \tilde{k}_\omega + 2a(e^a \cdot \mathbf{1}_{>0}) * \tilde{k}_\omega. \quad (3.8)$$

Proof. $\frac{p+a}{p-a} = 1 + \frac{2a}{p-a}$ is completely monotone on $p > a$, and $\tilde{K}_\omega$ on $p > b > a$; products of completely monotone functions are completely monotone. $B_b = 1 - \frac{2b}{p+\bar{b}}$ gives (3.8) by the convolution theorem. $\square$

3.4 Reading

The transfer is elementary on a window. Only $n < e^L$ contribute to k_ω on $[0, L]$, since $k_\omega^\Gamma(\cdot - \log n)$ vanishes below $\log n$. So $V_{\omega, L}$ is assembled exactly from $(2 \sinh x)^{\omega-1} e^{x/2}$, the numbers $\tilde{c}_n$ for $n < e^L$, and two exponentials, in precisely the way Q_L is assembled from (2.3). Section 5 does this.

The first-order kernel. As $\omega \downarrow 0$, off the atoms,

$$\frac{k_\omega(x)}{\omega} \longrightarrow 2n_\gamma(x) - 4 \cosh \frac{x}{2}, \quad \tilde{c}_n = 2\omega \Lambda(n) n^{-1/2} + O(\omega^2), \quad (3.9)$$

which is $-g_0$ of (2.11) off the atoms and at the atoms, as it must be by (2.9) ($\partial_\omega K_\omega|_0 = -a_0$). The continuous part is positive for $x < x_0$ and negative beyond, where $n_\gamma(x_0) = 2 \cosh \frac{x_0}{2}$, i.e. $u^3 + u^2 = 1$ with $u = e^{-x_0}$: $x_0 = 0.28120$. So k_ω is a positive integrable singularity at the origin, positive integrable singularities at every $\log n$, and a negative continuous background produced entirely by the rational factor.

Growth and fluctuation. $\hat{k}_\omega$ grows like $C_\omega e^{bx}$ with $C_\omega = (2\omega\Lambda(1+2\omega))^{-1} \rightarrow 1$, the residue of (3.7) at $p = b$: the pole of $\zeta(s - \omega)$ at $s - \omega = 1$, the prime number theorem's main term in the comb ($\sum_{n \leq N} \tilde{c}_n \sim N^b / (b\zeta(1+2\omega))$) smoothed by the Beta kernel. The exponential moving average at rate b maps e^{bx} to $\frac{1}{2}(e^{bx} - e^{-bx})$, so $g \mapsto g - 2\text{EMA}_b[g]$ sends Ce^{bx} to Ce^{-bx} : the subtraction kills the main term exactly and unconditionally. What remains is the fluctuation of $\hat{k}_\omega$ about its main term. Unconditionally $k_\omega(x) = O(e^{(\theta - \omega + \epsilon)x})$ with $\theta = \sup \text{Re}(\rho - \frac{1}{2})$; under RH, k_ω decays like $e^{-\omega x}$ with the residue expansion $\sum_\rho \frac{\xi(\rho - 2\omega)}{\xi'(\rho)} e^{(-\omega + i\gamma_\rho)x}$ [3, Section 5.2]. This is Proposition 2.3 at the level of the kernel.

What it does not say. It does not make V_ω a Markov operator, and cannot: a positive kernel of mass $K_\omega(0) = 1$ would make every compression a contraction and prove a zero-free strip. The first-order background is negative for $x > x_0$, and the sign changes at every ω computed [4]. Nor does structural positivity of $\hat{k}_\omega$ bound $\|V_{\omega,L}\|$: by Lemma 2.2, $V_{\omega,L} = (P_L B_b P_L)(P_L \hat{V}_\omega P_L)$, and the first factor is a contraction while the second has norm greater than one, so the factorization bounds $\|V_{\omega,L}\|$ by a number greater than one. The cancellation between the two factors is the whole content, as Proposition 2.3 says it must be. What the decomposition does is put the whole failure of positivity into one rational factor with explicit residues, and identify the positive part with objects that probability and geometry know how to produce — the subject of the next section.

4 What the pieces are, and what can realize them

4.1 The two proposals, stated precisely

Before naming the pieces we record what the two proposals of the introduction actually require, since the inherited note [4] found that both had been stated too strongly.

What is abelian. By Lemma 2.2 the map $H \mapsto P_L(h * \cdot)P_L$ is a homomorphism from the commutative algebra of causal symbols onto $\mathcal{A}_L$, so every $V_{\omega,L}$ and every $G_{\omega,L}$ lie in one commutative algebra, for all ω at fixed L . What is abelian is *translation invariance along the line*: each shift acts on I_L by a kernel depending only on the separation $x - y$, with the prime atoms at fixed separations, and two functions of the line momentum commute. Read as a condition on a realization, the insertions implementing the ω -deformation must act as functions of a single operator, uniformly along the line. That is a selection of a Cartan-like direction, not an impossibility.

The critical path is ordered. The endpoint generator $B_L = \frac{1}{2}\delta_{L/2} \otimes \text{ev}_{L/2}$ of the length direction satisfies $G_{\omega,L}B_L = 0 \neq B_L G_{\omega,L}$ [2, Proposition 9.3]. Along a path $t \mapsto (\omega(t), L(t))$ the transport $\mathcal{P} \exp \int (\dot{L}B_L - \dot{\omega}G_{\omega,L}) dt$ is a genuinely ordered product; only its value is path-independent, because the connection is flat.

Flatness is a tautology. If $V(\alpha)$ is any smooth family with $dV = \mathcal{A}V$ and dense range, then $0 = d^2V = (d\mathcal{A} - \mathcal{A} \wedge \mathcal{A})V$ forces the curvature to vanish. In the non-Abelian Stokes variation of an open line $W[C]$, the field strength enters as the *coefficient* $\mathcal{A}$ of the bulk deformation, and $\mathcal{A}$'s curvature on deformation space is zero by the same argument. So flatness constrains no realization; the object to match is the coefficient $G_{\omega,L}$ itself, together with the one-sided commutation with the endpoint term. What remains as the obstruction is the dichotomy: any realization must be exactly critical.

The Loewner affinities. Whole-plane Loewner evolution grows a hull K_t from 0 with logarithmic capacity t , so Loewner time is log-conformal-radius, the arithmetic coordinate $x = \log r$; the stabilizer of the marked points $0, \infty$ is the group of dilations and inversions, the exact symmetry group of the Weil form; the exterior Loewner equation $\partial_t g_t(z) = g_t(z) \frac{e^{iU_t} + g_t(z)}{e^{iU_t} - g_t(z)}$ is a tip equation with one scalar coefficient, like the endpoint variation $\partial_L V_{\omega,L} = B_L V_{\omega,L}$ [2, Proposition 9.1]; the chain is a path-ordered flow of Witt-algebra vector fields [13, 12]; and composition with a univalent self-map is a contraction on Hardy-type spaces by Littlewood subordination [14]. The proposal is therefore well motivated, and the question it poses is concrete: can such a chain produce the kernel k_ω , or at least the positive part $\tilde{k}_\omega$?

4.2 The archimedean factor is Bessel additivity

Proposition 4.1 (Bessel additivity). *Let Z_δ denote independent $\text{Gamma}(\frac{\delta}{2})$ variables — the value at any fixed time, up to the common factor $2t$, of a squared Bessel process of dimension δ started at 0. Then, with $U = Z_a/(Z_a + Z_{2\omega})$,*

$$U \sim \text{Beta}\left(\frac{a}{2}, \omega\right), \quad K_\omega^\Gamma(p) = \pi^\omega \frac{\Gamma(\frac{a}{2})}{\Gamma(\frac{b}{2})} \mathbb{E}[U^{p/2}], \quad Z_a + Z_{2\omega} \stackrel{d}{=} Z_b. \quad (4.1)$$

The archimedean delay is $x = -\frac{1}{2} \log U$, i.e. $(r_1/r_2)^2 = U$ in the ray variable $r = e^x$.

Proof. $G_\alpha/(G_\alpha + G_\beta) \sim \text{Beta}(\alpha, \beta)$ and $G_\alpha + G_\beta \sim G_{\alpha+\beta}$ for independent Gamma variables are classical; $\frac{a}{2} + \omega = \frac{b}{2}$. The additivity of squared Bessel processes in the dimension is Shiga–Watanabe [10, 11]. The moment identity is (3.6). $\square$

Three remarks. The two Bessel dimensions a, b are the two exponents e^{ax}, e^{-bx} of the rational factor: one set of numbers, $\frac{1}{2} \pm \omega$, because both come from $s \pm \omega$. The dimensions are fractional and below one, so there is no sphere and no angle; the Beta variable is the additivity fraction, and the archimedean “quarter” of the delay test [2, Section 11] is $\frac{a}{2}$ at $\omega = 0$, half a Bessel dimension. And the statement is about Bessel processes, which drive radial Loewner chains but are not chains: whether a conformal-radius or hitting law of a radial chain driven by a Bessel process of dimension b has U as a squared-radius ratio is open (Section 6).

4.3 A Loewner chain cannot produce the comb

There are two ways to identify the arithmetic coordinate with a Loewner variable, and neither produces the comb.

Lemma 4.2 (Rotation averages of composition operators are dilations). *Let ϕ be analytic near 0 with $\phi(0) = 0$ and let h be analytic. Then*

$$\frac{1}{2\pi} \int_0^{2\pi} h(e^{i\theta} \phi(e^{-i\theta} z)) d\theta = h(\phi'(0)z). \quad (4.2)$$

Consequently, for a random conformal map g with rotation-invariant law and deterministic $g'(0)$, $\mathbb{E}[h \circ g] = h(g'(0) \cdot)$ for every analytic h ; the same holds at ∞ for exterior maps $g(z) = e^{-t}z + O(1)$.

Proof. It suffices to take $h = z^n$. Then $e^{in\theta} \phi(e^{-i\theta} z)^n = e^{in\theta} \sum_{k \geq n} b_k e^{-ik\theta} z^k$ and the average keeps only $k = n$, with coefficient $b_n = \phi'(0)^n$. $\square$

So if the arithmetic coordinate is Loewner time, the expected composition operator of a rotation-invariant chain with the capacity normalization acts on H^2 , on the Dirichlet space, on any space of analytic functions, as $h \mapsto h(e^{-t}\cdot)$: a pure delay by t , symbol e^{-tp} , carrying no information about the chain. The shape of the curve lives in non-analytic observables — $|g|$, harmonic measure, $\mathbb{E}|g'|^q$ — and a Loewner realization of the transfer would have to act on those. The natural one is the radial process $\log |g_t(z)|$, and for it the other identification applies.

Proposition 4.3 (Far-field transport is a pure delay). *Let K_t be a whole-plane Loewner hull of logarithmic capacity t , deterministic or random, with any number of tips, and $g_t : \hat{\mathbb{C}} \setminus K_t \rightarrow \hat{\mathbb{C}} \setminus \mathbb{D}$ its normalized map, $g_t(z) = e^{-t}z + O(1)$. Then $\log |g_t(z)| = \log |z| - t + O(|z|^{-1})$ uniformly in $\arg z$. Hence if the induced transition kernel $P_t(y, dy')$ from $y = \log |z|$ to $y' = \log |g_t(z)|$, averaged over $\arg z$ and over the chain's law, is translation-invariant in y , it equals $\delta_{y'=y-t}$. In particular no Loewner chain induces a translation-invariant radial kernel with atoms at $y' = y - \log n$, $n \geq 2$.*

Proof. The normalization at infinity is the definition of the capacity parametrization; translation invariance forces $P_t(y, \cdot)$ to equal its $y \rightarrow \infty$ limit. $\square$

Away from the far field the kernel depends on the distance to the hull and is not translation-invariant; and in either case $t \mapsto g_t(z)$ is continuous even for a càdlàg driving function, since the Loewner equation is an ODE whose field is smooth in g away from the circle, so a jump of the driving process moves the tip, not the transported points. Echoes at the fixed separations $\log n$ with position-independent weights are not a Loewner phenomenon in any reading. What Loewner or Bessel processes can supply is the archimedean factor alone.

4.4 What the comb is

On functions of the arithmetic coordinate let $\mu_n f(x) = f(x - \log n)$. The μ_n are isometries of $L^2(\mathbb{R}, dx)$ with $\mu_n \mu_m = \mu_{nm}$, and the comb operator is

$$C_\omega = \sum_{n \geq 1} \tilde{c}_n \mu_n, \quad \tilde{c}_n = n^{-1/2} (n^\omega * \mu(n) n^{-\omega}) = n^{\omega - \frac{1}{2}} \prod_{p|n} (1 - p^{-2\omega}), \quad (4.3)$$

a Dirichlet series in the multiplicative semigroup $\mathbb{N}^\times$ acting on the ray by dilations: the image of $\zeta(s - \omega)/\zeta(s + \omega)$ under the representation $n^{-s} \mapsto \mu_n$, with the $n^{-1/2}$ making it unitary on $L^2(dx)$. This is the Hecke algebra of $\mathbb{N}^\times$ in the Bost–Connes system [15], which [2, Section 10.4] had already named as the structure “invariant under $r \mapsto \lambda r$ but distinguishing the ratios p^m ” that the prime atoms require; the decomposition says exactly which element of that algebra the transfer contains, with which coefficients, at each shift. Its local factor at p , $\frac{1-p^{-\omega}z_p}{1-p^\omega z_p}$ with $z_p = e^{-s \log p}$, is a one-delay feedback loop of length $\log p$ with loop gain p^ω : its kernel before the $n^{-1/2}$ is $\delta_0 + (1 - p^{-2\omega}) \sum_{k \geq 1} p^{k\omega} \delta_{k \log p}$, and a traversal of the loop multiplies by the scalar p^ω , which is the abelian monodromy at the puncture of [2, Section 9.4], made explicit. To first order in ω the coefficients are $2\omega \Lambda(n) n^{-1/2}$, the von Mangoldt function, and at second order the composite weights appear: $\tilde{c}_6 = \omega^2 \cdot 4 \log 2 \log 3 / \sqrt{6} + O(\omega^3)$.

4.5 The endpoint $\omega = \frac{1}{2}$ is the modular surface

At $\omega = \frac{1}{2}$: $a = 0$, $b = 1$, $\frac{p+a}{p-a} = 1$, so

$$\widehat{K}_{1/2}(p) = \widetilde{K}_{1/2}(p) = \frac{\Lambda(p)}{\Lambda(p+1)} = \varphi\left(\frac{p+1}{2}\right), \quad \varphi(s') = \sqrt{\pi} \frac{\Gamma(s' - \frac{1}{2}) \zeta(2s' - 1)}{\Gamma(s') \zeta(2s')} = \frac{\Lambda(2s' - 1)}{\Lambda(2s')}, \quad (4.4)$$

where φ is the scattering matrix of the Eisenstein series of $PSL(2, \mathbb{Z}) \backslash \mathbb{H}$ [17], the coefficient of $y^{1-s'}$ in the constant term of $E(z, s')$; its pole at $s' = 1$ is $p = b = 1$. The comb weights are $\tilde{c}_n = \prod_{p|n} (1 - p^{-1}) = \varphi_{\text{Euler}}(n)/n$, the density of reduced fractions with denominator n — the count of cosets in the cusp's Fourier expansion that produces $\zeta(2s' - 1)/\zeta(2s')$ in the first place. The Beta part degenerates: $\text{Beta}(0, \frac{1}{2})$, and $k_{1/2}^\Gamma(x) = 2(1 - e^{-2x})^{-1/2}$ does not decay, which is the pole of $K_{1/2}^\Gamma$ at $p = 0$. And

$$K_{1/2} = B_1 \varphi\left(\frac{p+1}{2}\right) = \frac{p-1}{p+1} \varphi\left(\frac{p+1}{2}\right) : \quad (4.5)$$

the transfer at the endpoint is the Eisenstein scattering matrix with its pole at $s' = 1$ removed by one Blaschke factor, which is what Lax and Phillips do when they pass to the orthocomplement of the constant function [16] (a reading; the identity is exact). So the Markov part has, at the endpoint, a classical geometric realization in which the comb is a coset sum and the correction is the removal of the residual spectrum. For $\omega < \frac{1}{2}$ Eisenstein constant terms give only unit offsets in $2s'$, so the interior of the family is not realized by any congruence-subgroup scattering matrix; this was already observed in [3]. All of (4.4)–(4.5) and the totient identity are checked numerically in Appendix A.

4.6 Summary of the dictionary

Piece	Symbol	Realization
Archimedean delay	$K_\omega^\Gamma = \pi^\omega \Gamma(\frac{s-\omega}{2})/\Gamma(\frac{s+\omega}{2})$	Bessel additivity in dimensions a , $2\omega \rightarrow b$ (Prop. 4.1)
Comb	$K_\omega^\zeta = \zeta(s-\omega)/\zeta(s+\omega)$	Hecke operator $\sum \tilde{c}_n \mu_n$ of $\mathbb{N}^\times$; not Loewner (Prop. 4.3)
Correction	$R_\omega = B_b/B_a$; cleanly B_b	Forced by unimodularity (Prop. 3.4); no arithmetic
Endpoint $\omega = \frac{1}{2}$	$K_{1/2} = B_1 \varphi$	Modular surface; Lax–Phillips (§4.5)
The whole	$K_\omega = B_b \hat{K}_\omega$	Positive measure minus twice its EMA (Prop. 3.5)

5 The contraction margin at the first horizon

The decomposition makes $V_{\omega,L}$ computable from elementary functions. This section builds it, says what its contraction data must be to second order in ω , and reports the computation at $L = \log 3$, the first horizon at which the margin m_L of the Weil form is recorded [7]. The programme is described in Appendix A; it is exploratory (mpmath, 40 digits) and every number has a record beside it.

5.1 The assembly

In the orthonormal sine basis $e_j(x) = \sqrt{2/L} \sin(j\pi(x + \frac{L}{2})/L)$, $j = 1, \dots, N$, on I_L , the Galerkin matrix of the compression is

$$V_{jk} = \langle e_j, V_{\omega,L} e_k \rangle = \int_0^L k_\omega(t) S_{jk}(t) dt, \quad S_{jk}(t) = \int_{-L/2+t}^{L/2} e_j(x) e_k(x-t) dx, \quad (5.1)$$

because $V_{\omega,L}$ is a causal convolution. The one-sided autocorrelations are elementary: with $\alpha = \pi/L$ and $\sigma = (-1)^{j+k}$,

$$\begin{aligned} S_{jj}(t) &= \frac{(L-t) \cos(j\alpha t) + \sin(j\alpha t)/(j\alpha)}{L}, \\ S_{jk}(t) &= \frac{1}{L} \left[\frac{\sigma \sin(k\alpha t) - \sin(j\alpha t)}{(j-k)\alpha} + \frac{\sigma \sin(k\alpha t) + \sin(j\alpha t)}{(j+k)\alpha} \right] \quad (j \neq k). \end{aligned} \quad (5.2)$$

The kernel is assembled in the form of Proposition 3.1: $k_\omega = \sum_{n < e^L} \tilde{c}_n \kappa_\omega(\cdot - \log n)$ with $\kappa_\omega = k_\omega^\Gamma - 4\omega b(e^{-b} \mathbf{1}_{>0}) * k_\omega^\Gamma - 4\omega a(e^a \mathbf{1}_{>0}) * k_\omega^\Gamma$. At $L = \log 3$ the atoms are $n = 1, 2$. The t -integral is the only quadrature. On each interval between consecutive atoms the integrand is written in the local coordinate $\tau = t - \log n_0$ and the singularity $\tau^{\omega-1}$ is removed by the substitution $\tau = u^{1/\omega}$, after which a tanh-sinh rule with step h is applied. The local coordinate is essential: the spike carries half its mass below $\tau = 2^{-1/\omega}$, and forming $t = \log n_0 + \tau$ in finite precision and subtracting again destroys it (the first version of the programme reported $\|V\| > 1$ for exactly this reason).

Three quantities are then read from the $N \times N$ matrix: $\|V_N\|$, which must be below one; $\lambda_{\min}(D_N)$ with $D_N = I - V_N^\top V_N$, compared through (1.3) with $m_L^{(N)}$, the smallest eigenvalue of Q_L in the same basis, computed in closed form from (2.3) by the assembler of [8]; and the first-order law on the fixed vector e_1 , where $Q_L[e_1] = 4.3474 \times 10^{-4}$ is 7000 times m_L and the check is insensitive to the noise floor.

5.2 What the numbers should be

Write $G = G_{0,L}$ for the compressed generator at $\omega = 0$ and split it into Hermitian and skew parts, $G = Q + A$, so that $Q = Q_L$ is the Weil form and A is a Hilbert transform on the interval together with the odd part of the comb and smooth terms.

Proposition 5.1 (The defect to second order). *On the compression,*

$$D_{\omega,L} = 2\omega Q - \omega^2(2Q^2 + [Q, A]) + O(\omega^3), \quad (5.3)$$

and if f is a normalized eigenvector of Q with eigenvalue m then $\langle f, D_{\omega,L} f \rangle / 2\omega = m - \omega m^2 + O(\omega^2)$. In particular $\lambda_{\min}(D_{\omega,L}) / 2\omega = m_L - \omega m_L^2 + O(\omega^2)$: the relative correction to the first-order law at the minimizer is $-\omega m_L$ and nothing else.

Proof. By (2.9), $K_\omega = \exp(-\int_0^\omega a_{\omega'} d\omega')$, and a_ω is even in ω , so $K_\omega = \exp(-\omega a_0 + O(\omega^3))$ as symbols on $\text{Re } p > b$, hence as causal kernels. By Lemma 2.2 compression is a homomorphism, so $V_{\omega,L} = \exp(-\omega G + O(\omega^3)) = I - \omega G + \frac{\omega^2}{2} G^2 + O(\omega^3)$ and $V^* V = I - \omega(G + G^*) + \omega^2(G^* G + \frac{1}{2} G^2 + \frac{1}{2} G^{*2}) + O(\omega^3)$. With $G = Q + A$: $G^* G = Q^2 + QA - AQ - A^2$ and $\frac{1}{2}(G^2 + G^{*2}) = Q^2 + A^2$, so the bracket is $2Q^2 + [Q, A]$. For the eigenvector, $\langle f, [Q, A] f \rangle = \langle Qf, Af \rangle - \langle A^* f, Qf \rangle = 2 \text{Re} \langle Qf, Af \rangle = 2m \text{Re} \langle f, Af \rangle = 0$ since $A^* = -A$, and $\langle f, Q^2 f \rangle = m^2$. The eigenvalue statement is first-order perturbation theory for the lowest eigenvalue of Q , which is an eigenvalue with a smooth eigenvector (the form is of logarithmic order on the interval, bounded below with compact resolvent) and is taken simple, as the computations show. The expansions are of matrix elements $\int_0^L k_\omega(t) S(t) dt$ against smooth S , which are analytic in ω : the spike integrates against a constant to ϵ^ω . $\square$

Proposition 5.2 (The Galerkin defect). *Let E_N be the span of the first N basis functions, P_N its projection, $V_N = P_N V_{\omega,L} P_N$, and f_N the normalized minimizer of Q on E_N with eigenvalue $m_L^{(N)}$. Then*

$$D_N := I_N - V_N^\top V_N = P_N D_{\omega,L} P_N + P_N V^* (I - P_N) V P_N \geq P_N D_{\omega,L} P_N, \quad (5.4)$$

and

$$\frac{\langle f_N, D_N f_N \rangle}{2\omega} = m_L^{(N)} - \omega (m_L^{(N)})^2 + \frac{\omega}{2} \left(\|(I - P_N) A f_N\|^2 - \|(I - P_N) Q f_N\|^2 \right) + O(\omega^2). \quad (5.5)$$

If the defect is instead computed nested in $E_{N'} \supset E_N$, $D_{N \subset N'} := I_N - (V_{N'}^\top V_{N'})_{N \times N} = P_N (I - V^* P_{N'} V) P_N$, the term $\frac{\omega}{2} \|(I - P_N) G f_N\|^2$ hidden in (5.5) is replaced by $\frac{\omega}{2} \|(I - P_{N'}) G f_N\|^2$, which tends to zero as $N' \rightarrow \infty$.

Proof. $V_N^\top V_N = P_N V^* P_N V P_N$ and $P_N \leq I$ give the identity and the inequality. On $f_N \in E_N$, $(I - P_N)Vf_N = -\omega(I - P_N)Gf_N + O(\omega^2)$, so the extra term is $\omega^2\|(I - P_N)Gf_N\|^2 + O(\omega^3)$. In Proposition 5.1 evaluated on f_N , which is an eigenvector of $P_N Q P_N$ but not of Q , write $Qf_N = m_L^{(N)} f_N + (I - P_N)Qf_N$; then $\|Qf_N\|^2 = (m_L^{(N)})^2 + \|(I - P_N)Qf_N\|^2$ and $\text{Re}\langle Qf_N, Af_N \rangle = \text{Re}\langle (I - P_N)Qf_N, (I - P_N)Af_N \rangle$, since $\text{Re}\langle f_N, Af_N \rangle = 0$. Collecting these with $\|(I - P_N)(Q + A)f_N\|^2$ expanded gives (5.5). $\square$

So a finite-basis computation must show a *positive* first-order excess over $m_L^{(N)}$, dominated by the energy of Af_N outside the basis. Af_N has logarithmic endpoint singularities from the Hilbert transform and nonzero endpoint values from the shifted copies, so its sine coefficients decay like $1/n$ and the excess should fall like $1/N'$ under nesting. Both predictions are tested below.

Proposition 5.3 (The first-order tail). *For $L > 0$, integrating over $[0, L]$,*

$$\lim_{\omega \downarrow 0} \frac{1}{\omega} \int_L^\infty k_\omega(t) dt = 8 \sinh \frac{L}{2} - 2 \sum_{n < e^L} \frac{\Lambda(n)}{\sqrt{n}} - \log 2\pi - \gamma_E - \int_0^L \left(2n_\gamma(u) - \frac{1}{u} \right) du - \log L. \quad (5.6)$$

At $L = \log 3$ the five terms are 4.618802, -0.980258 , -1.837877 , -0.577216 , -0.592935 , total 0.630516.

Proof. $\int_0^\infty k_\omega = K_\omega(0) = \xi(\frac{1}{2} - \omega)/\xi(\frac{1}{2} + \omega) = 1$ for every ω , so the tail is minus the head, and $\partial_\omega|_0 \int_0^L k_\omega = \int_0^L k_1$ with $k_1 = -g_0$ of (2.11). The pole terms integrate to $-2 \int_0^L (e^{-t/2} + e^{t/2}) dt = -8 \sinh \frac{L}{2}$, the constant to $\log \pi$, the comb to $2 \sum_{n < e^L} \Lambda(n)n^{-1/2}$, and the finite part, by (2.12) with $\phi = \mathbf{1}_{[0,L]}$ and $E_1(x) = -\gamma_E - \log x + O(x)$, to $\gamma_E + \log 2 + \int_0^L (2n_\gamma - \frac{1}{u}) du + \log L$. Collect and negate. $\square$

This checks the mass of the assembled kernel by a formula that integrates the poles, the digamma and the primes directly, where the kernel carries them through the Beta convolution and two exponential smoothings; it uses no zeros.

5.3 Results

All runs: $L = \log 3$ exactly, atoms $n \in \{1, 2\}$, 40 digits, tanh-sinh step $1/h = 16$ unless stated. In the same basis, $m_L^{(16)} = 6.84655 \times 10^{-8}$, $m_L^{(24)} = 6.24751 \times 10^{-8}$, $m_L^{(32)} = 6.21505 \times 10^{-8}$ (even block); the converged value is 5.66×10^{-8} [8] (recorded 5.54×10^{-8} [7]), so the sine basis is far from converged at these N , and every comparison is *within a basis*, which is what Proposition 5.2 is about.

Table 1: The ω -sweep at $N = 24$. The last column is $(1 - \|P_N V e_1\|^2)/(2\omega Q_L[e_1])$.

ω	$1 - \ V_N\ $	$\lambda_{\min}(D_N)$	$\lambda_{\min}(D_N)/2\omega$	ratio to $m_L^{(24)}$	e_1 ratio
0.002 ($1/h = 20$)	1.2460×10^{-10}	2.49192×10^{-10}	6.22980×10^{-8}	0.99717	0.99883
0.01	6.2543×10^{-10}	1.25087×10^{-9}	6.25435×10^{-8}	1.00109	0.99465
0.02	1.2606×10^{-9}	2.52117×10^{-9}	6.30292×10^{-8}	1.00887	0.99057
0.05	3.2101×10^{-9}	6.42021×10^{-9}	6.42021×10^{-8}	1.02764	0.98581
0.1	6.6085×10^{-9}	1.32171×10^{-8}	6.60854×10^{-8}	1.05779	1.00203

Contraction and the first-order law. V_N is a strict contraction at every ω (Table 1), with $1 - \|V_N\| = \frac{1}{2}\lambda_{\min}(D_N)$ to all printed digits as it must be for $\lambda_{\min} \ll 1$, and $\lambda_{\min}(D_N)/2\omega \rightarrow m_L^{(24)}$ as $\omega \downarrow 0$. The transfer side, assembled from k_ω^Γ , $\tilde{c}_n$ and R_ω , and the form side, assembled from (2.3), agree at first order in ω to the precision of the quadrature. On e_1 , which is not an eigenvector, the correction is genuinely of order ω , as (5.3) predicts for a vector on which $\langle Qe_1, Ae_1 \rangle \neq 0$.

The quadrature floor. The step $1/h = 16$ resolves $\lambda_{\min}(D_N)$ to about 6×10^{-12} : this is 2.5% of the signal at $\omega = 0.002$ (where the $1/h = 16$ run gave ratio 0.97507 and the $1/h = 20$ run 0.99717) and $\pm 0.5\%$ at $\omega = 0.01$ (where $N = 24$ and $N = 32$ gave deviations of opposite sign, $+0.11\%$ and -0.50%). At $1/h = 20$ the floor is below 10^{-12} ; at $1/h = 12$ it exceeds the signal at $\omega = 0.002$. The mass $\int_0^L k_\omega$ agrees between $1/h = 16$ and 20 to 10^{-15} , so the floor is in the oscillatory matrix elements, not the mass.

Table 2: Basis size and the nested defect at $\omega = 0.1$ (floor 0.05% of the signal), and one row at $\omega = 0.05$.

ω	N	N' (nested)	ratio to $m_L^{(N)}$	excess
0.1	16	–	1.06508	6.5%
0.1	24	–	1.05779	5.8%
0.1	32	–	1.04540	4.5%
0.1	24	40	1.03792	3.8%
0.1	24	64	1.02035	2.0%
0.05	24	–	1.02764	2.8%
0.05	24	64	1.00612	0.6%

The second-order behaviour. The excess over $m_L^{(N)}$ at $\omega \geq 0.02$ is positive and close to linear in ω (relative slopes 0.44, 0.55, 0.58 at $\omega = 0.02, 0.05, 0.1$), which is the Galerkin term of Proposition 5.2, not the $-\omega m_L$ of Proposition 5.1. Table 2 tests the prediction: the excess decreases with N , and under nesting at fixed $N = 24$ falls as $5.8 \rightarrow 3.8 \rightarrow 2.0\%$ for $N' = 24, 40, 64$, against $5.8 \times 24/N' = 5.8, 3.5, 2.2\%$ for a $1/N'$ law. Fitting the nested-64 excesses at $\omega = 0.05, 0.1$ to $\alpha\omega + \beta\omega^2$ gives $\alpha \approx 0.04$ and $\beta \approx 1.6$: the linear artefact is essentially gone at $N' = 64$, and what remains is $\lambda_{\min}(D)/2\omega \approx m_L(1 + 1.6\omega^2)$, a second-order coefficient of order unity relative to m_L , where the a-priori estimate $O(\omega^2 \kappa_L^2)$ with $\kappa_{\log 3} = 61$ [7] allows 3.7×10^3 . The same trend appears on e_1 : its ratio at $\omega = 0.1$ is 1.0273, 1.0020, 0.9876 for $N = 16, 24, 32$, crossing below one as the truncation term dies and the genuine $-\omega(\dots)$ term remains.

Table 3: The mass on the window. Proposition 5.3 gives 0.630516 for the $\omega \rightarrow 0$ limit of the last column; linear extrapolation of the first two rows gives 0.63052.

ω	$\int_0^L k_\omega$	$(1 - \int_0^L k_\omega)/\omega$
0.002	0.9987404683	0.629766
0.01	0.9937323187	0.626768
0.02	0.9875394433	0.623028
0.05	0.9694073969	0.611852
0.1	0.9406626582	0.593373

The mass. Table 3 against Proposition 5.3: the assembled kernel — Beta convolution, comb, two exponential smoothings, local coordinates, substitution, tanh–sinh — has the right first-order mass beyond the horizon to five digits, by a formula that knows nothing of the assembly.

5.4 Reading

The elementary assembly is validated end to end. The kernel side and the form side agree at the first-order law to the precision of the quadrature; their second-order disagreement is the one Proposition 5.2 prescribes and dies as prescribed; and the mass agrees with an independent closed form. There is now a working instrument that computes $\|V_{\omega,L}\|$, $\lambda_{\min}(D_{\omega,L})$ and their ω -dependence from the integers below e^L and elementary functions.

What it says at $L = \log 3$ is what Proposition 2.3 says it must: contraction, with margin ωm_L to first order and a positive second-order correction. Nothing about the zeros is learned at one horizon where $m_L > 0$ is already known; the dichotomy is about all horizons, and the margins at $\log 5, \log 7$ are $10^{-17}, 10^{-27}$, beyond what a double-precision registered check can resolve and within what this instrument can at 40 digits, at a cost that grows with the number of atoms rather than with L .

Proposition 5.1 is worth keeping for its own sake: the defect of the compressed all-pass is, to second order, $2\omega Q - \omega^2(2Q^2 + [Q, A])$, so the only first-order correction to the first-order law at the minimizer is $-\omega m_L^2$, and the commutator of the Weil form with its skew partner — the compressed Hilbert transform of the interval — governs the second-order behaviour away from the minimizer.

6 What remains, ranked

The decomposition closes some questions and opens others. We list what is closed first, so that it is not redone, and then what is open, in the order we would take it.

6.1 Closed

1. The comb is not a Loewner phenomenon, in either identification of the variables, for single- or multi-tip chains, deterministic or Lévy-driven (Section 4.3).
2. The rational correction carries no arithmetic and its residues are not data (Proposition 3.4); the inherited note’s expectation that a realization would fail “at two numbers” is withdrawn.
3. Structural positivity of the Markov part bounds nothing about $\|V_{\omega,L}\|$ (Section 3); the cancellation between the positive part and its EMA is the content.
4. The far-right-line Fourier realization of the transfer is superseded for numerical purposes by the elementary assembly (Section 5.1).

6.2 Open

1. **Finish the contraction side.** A standard-library registered version of the assembly at $L = \log 3$, the one recorded horizon where $2\omega m_L \approx 10^{-9}$ is resolvable in double precision, asserting contraction, $\lambda_{\min}(D_N)/2\omega = m_L^{(N)}(1 + \delta)$ with $|\delta| < 1\%$ at $\omega = 0.01$, and the mass against Proposition 5.3. The horizons $\log 5, \log 7$ at 40 digits, exploratory. The cumulative Cayley coordinate of [2, Proposition 7.7] from the same matrices. And the phase-matched Gram control of [3] on the form side against the transfer’s margin: the transfer has no lattice analogue, so this control measures what the transfer knows that a lattice does not.

2. **The Lax–Phillips dictionary at $\omega = \frac{1}{2}$** , object by object: which subspace the compression P_L is in their energy form, what the flux identity (2.13) is there, and what their semigroup estimate says about the contraction margin at the endpoint. Then which of their objects the ω -deformation moves and which it cannot reach.
3. **The Beta law from a radial chain.** Whether a conformal-radius or hitting law of radial Loewner evolution driven by a Bessel process of dimension b has the additivity fraction of Proposition 4.1 as a squared radius ratio. If yes, the archimedean factor has a Loewner realization and its quarter is a Bessel dimension; if no, the archimedean factor is a Bessel-process statement and not a Loewner one, and the investigation’s name records its origin rather than a direction.
4. **The comb in the Bost–Connes algebra.** C_ω is the image of $\zeta(s - \omega)/\zeta(s + \omega)$ under $n^{-s} \mapsto \mu_n$. What the KMS structure of that algebra says about the compression $P_L C_\omega P_L$, whose norm grows like e^{bL} , and whether the subtraction 2EMA_b has a meaning there. A reading task with a small chance of a real statement.
5. **The criticality in filter terms.** The transfer is a lossless filter with impulse response $\widehat{k}_\omega - 2\text{EMA}_b[\widehat{k}_\omega]$, $\widehat{k}_\omega \geq 0$, and RH is its stability at every $\omega > 0$. A clean statement of why no passivity theorem for that class can avoid the location of the poles — what property of $\widehat{k}_\omega$ beyond positivity would be needed — would be the honest end of this line.

6.3 Status of every statement

Written proof, unconditional: Propositions 2.1, 3.1–3.5, 4.1, 4.3, 5.1–5.3, Lemmas 2.2 and 4.2, and the identities of Section 4.5. *Quoted from [2]:* Proposition 2.3 and the flow identities. *Labelled numerical computation (unregistered):* Section 5.3 and Appendix A. *Reading:* the Lax–Phillips remark of Section 4.5; the Bost–Connes identification beyond the algebra written out; the $1/N'$ law and the two-point fit of Section 5.3. *Not claimed:* a realization of the transfer, a contraction at any horizon beyond $\log 3$, a positivity statement, convergence of the sine basis, or anything about the zeros.

A Programmes, records, and numerical lessons

All programmes live in `numerics/exploratory/` of the investigation folder, use `mpmath`, and are unregistered in the sense of the program’s conventions: nothing replays them automatically, and every value is either an identity checked to working precision or a labelled quadrature. Nothing here is a positivity certificate.

decomposition_checks.py (30 digits, record `decomposition_checks_omega0.1.json`), at $\omega = 0.1$: complete monotonicity of $\widehat{K}_\omega$ through six derivatives at $p \in \{0.8, 1, 2\}$; $K_\omega = B_b \widehat{K}_\omega$ at five complex p to relative 2×10^{-31} ; $\widehat{K}_{1/2}(p) = \varphi(\frac{p+1}{2})$, $\widehat{K}_{1/2} = \widehat{K}_{1/2}$ and $K_{1/2} = B_1 \varphi$ to 10^{-31} , and $\tilde{c}_n|_{\omega=1/2} = \varphi_{\text{Euler}}(n)/n$ for $n < 200$ in exact rationals; (3.6) by Beta moments and by the Gamma-ratio form to 4×10^{-31} , with the Beta density pulled back to k_ω^Γ pointwise; the EMA identity (3.8) against the R_ω assembly at four points to 2×10^{-31} , with $\widehat{k}_\omega > 0$ there; and the first-order comb weights $\tilde{c}_n/(2\omega\Lambda(n)n^{-1/2})$ at $\omega = 0.01$ within 1.00001–1.00006 for $n = 2, 3, 5, 7$, and $\tilde{c}_6/\omega^2 = 1.2436$ against $4 \log 2 \log 3/\sqrt{6}$.

transfer_kernel.py (record `transfer_kernel_omega0.1.json`): the kernel k_ω assembled from k_ω^Γ , $\tilde{c}_n$ and R_ω , with checks of the Laplace transforms, the Dirichlet coefficients, $R_\omega \widetilde{K}_\omega = K_\omega$, complete monotonicity, the first-order law (3.9), a kernel table, and the truncated kernel transform against $K_\omega(p)$, which converges geometrically in the cutoff.

contraction_margin.py (40 digits; usage `omega L N [1/h] [Nbig]`, with L written as `log3` parsed exactly and atoms taken with $\log n < L$ strictly): the assembly of Section 5.1. The closed form (5.2) is checked against quadrature to 10^{-41} . The tanh-sinh rule uses nodes $u_k = \frac{U}{2}(1 + \tanh(\frac{\pi}{2} \sinh kh))$, $|k| \leq 4.5/h$, on each interval $(0, U)$, $U = (\log n_1 - \log n_0)^\omega$, after the substitution $\tau = u^{1/\omega}$; with $1/h = 16$ this is 262 kernel evaluations, about 40 seconds, the whole cost of a run. $m_L^{(N)}$ and $Q_L[e_1]$ come from `weil_sine_basis.py` of the Wilson-lines exploratory folder, in closed form. With a fifth argument $N' > N$ the matrix is assembled at N' and the nested defect of Proposition 5.2 is reported. Records `cm_om{omega}_log3_N{N}.json`, with suffixes `_h20` and `_nested{N'}` where used, for every row of Tables 1–3.

Lessons. Recorded so that the registered version does not rediscover them.

1. *Local coordinates.* The spike $k_\omega^\Gamma(\tau) \sim \omega \pi^\omega 2^\omega \tau^{\omega-1}$ puts half its mass below $\tau = 2^{-1/\omega}$. Any formulation that forms the global time $t = \log n_0 + \tau$ and recovers τ by subtraction loses it at every finite precision; the kernel must be evaluated as a function of τ . Symptom: $\|V\| > 1$ by 10^{-3} and a first-order check off by a factor 100.
2. *The substitution and the rule.* $\tau = u^{1/\omega}$ removes the singularity exactly; after it a tanh-sinh rule converges doubly exponentially, but the step needed grows as ω shrinks. $1/h = 16$ gives a floor of 6×10^{-12} in $\lambda_{\min}(D)$, $1/h = 20$ below 10^{-12} ; $1/h = 12$ is insufficient at $\omega \leq 0.002$. Write the rule out; a library's node generator may not follow its documentation.
3. *Atoms at the horizon.* With L entered as a decimal slightly above $\log 3$, or formed at default precision before the working precision is set, the atom $n = 3$ enters with a sliver $[\log 3, L]$ of length 10^{-17} and carries $(10^{-17})^\omega$ of its spike mass — 68% of it at $\omega = 0.01$. This changes $\int_0^L k_\omega$ (to 1.0027 from 0.9937) but not the operator, because $S_{jk}(L) = 0$ for all j, k . Parse $L = \log n$ exactly, set the precision first, take $\log n < L$ strictly. The record `cm_om0.01_log3_N16_h12_edgeatom_stale.json` is that run, kept as the example.
4. *Sign of the Galerkin bias.* $D_N \geq P_N D_{\omega, L} P_N$: the plain Galerkin defect overestimates at first order in ω by the energy of Af_N outside the basis; the nested defect removes most of it at the cost of assembling V at N' . Compare within a basis, never a Galerkin $\lambda_{\min}(D_N)$ against a converged m_L .

B Provenance, and the use of language models

This manuscript was drafted for Edward Baker by Claude Fable 5.1 (Anthropic, model `claude-fable-5-1`), at his direction and in dialogue with him, on 17 September 2026, as the first consolidated report of the Loewner investigation of the SUSY-positivity program. The investigation was opened by Edward Baker from a research note of the Wilson-lines investigation [4]; the questions it pursues — what exactly is abelian in the shift flow, whether flatness excludes curved realizations, and whether a Loewner-type evolution could construct the deformation paths independently — are his, and the answers here were developed in response to them. The

mathematical content consolidates three research notes of the program [4, 5, 6], all written by the same model with the model named at their head, and the numerical programmes and records cited in Appendix A, also written by it. The proofs are short and are meant to be checked line by line by a human reader; the numerical statements are reproducible from the programmes and records in the repository; the readings are labelled as such. Nothing in this document should be cited as established beyond what its own proofs establish.

References

- [1] Shared background of this program, *SUSY positivity research program: mathematical background*, 11 September 2026, expanded 13 and 17 September 2026, `papers/susy-positivity/background.pdf`.
- [2] Companion manuscript of this program, *Deformation flows for the shifted Weil family: the transfer as an all-pass filter*, working draft 0.8, 17 September 2026, `papers/susy-positivity/investigations/wilson-lines`. Results are cited by their numbers in that draft.
- [3] Review of the Wilson-lines investigation, 17 September 2026, `investigations/wilson-lines/reviews/review_claude-fable-5-1_2026-09-17.md`.
- [4] Research note of this program, *Loewner evolution and the Markov decomposition of the transfer*, 17 September 2026, `investigations/wilson-lines/notes/LOEWNER_AND_THE_MARKOV_DECOMPOSITION_20260917.md`.
- [5] Research note of this program, *The Markov part of the transfer and what can realize it: opening note*, 17 September 2026, `investigations/loewner/notes/MARKOV_PART_AND_REALIZATIONS_20260917.md`.
- [6] Research note of this program, *The contraction margin at the first horizon: the elementary assembly validated end to end*, 17 September 2026, `investigations/loewner/notes/CONTRACTION_MARGIN_FIRST_RUN_20260917.md`.
- [7] Research note of this program, *The critical path in the shift-length plane*, 17 September 2026, `investigations/wilson-lines/notes/CRITICAL_PATH_20260917.md`.
- [8] Research note of this program, *The frame bound and the density*, 17 September 2026, `investigations/wilson-lines/notes/FRAME_BOUND_AND_THE_DENSITY_20260917.md`.
- [9] R. L. Schilling, R. Song and Z. Vondraček, *Bernstein Functions: Theory and Applications*, 2nd ed., de Gruyter Studies in Mathematics 37, de Gruyter, Berlin, 2012.
- [10] T. Shiga and S. Watanabe, *Bessel diffusions as a one-parameter family of diffusion processes*, *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* **27** (1973), 37–46.
- [11] D. Revuz and M. Yor, *Continuous Martingales and Brownian Motion*, 3rd ed., Grundlehren der mathematischen Wissenschaften 293, Springer, Berlin, 1999, Chapter XI.
- [12] G. F. Lawler, *Conformally Invariant Processes in the Plane*, Mathematical Surveys and Monographs 114, American Mathematical Society, Providence, 2005.
- [13] M. Bauer and D. Bernard, *Conformal field theories of stochastic Loewner evolutions*, *Communications in Mathematical Physics* **239** (2003), 493–521, [arXiv:hep-th/0210015](https://arxiv.org/abs/hep-th/0210015).
- [14] J. H. Shapiro, *Composition Operators and Classical Function Theory*, Universitext, Springer, New York, 1993.
- [15] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, *Selecta Mathematica (N.S.)* **1** (1995), 411–457.

- [16] P. D. Lax and R. S. Phillips, *Scattering Theory for Automorphic Functions*, Annals of Mathematics Studies 87, Princeton University Press, Princeton, 1976.
- [17] H. Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., Graduate Studies in Mathematics 53, American Mathematical Society, Providence, 2002.